

Fine-tuning Small Language Models for Cross-Prompt Arabic Essay Scoring: A Benchmarking Study

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Abstract

Automated Essay Scoring (AES) has been studied since the early days of computational methods. While recent advances incorporating large language models (LLMs) have significantly improved performance, most research has focused on English. In this paper, we present the first benchmark of Arabic cross-prompt AES using Arabic-centric LLMs. We evaluate ALLAM 7B, JAIS 8B, and FANAR 9B under zero-shot, few-shot, and fine-tuning settings, and compare them against established AES systems and frontier LLMs. Fine-tuning substantially improves performance across all models, with FANAR achieving state-of-the-art results and outperforming the strongest baseline by 8 points on LAILA dataset. We further show that fine-tuned Arabic-centric LLMs generalize effectively in cross-dataset setting (LAILA $\rightarrow$ ZAE-BUC). Finally, we analyze the best-performing model, examining error patterns, trait interactions, and the effects of multi-trait autoregressive scoring.

1 Introduction

Automated Essay Scoring (AES) is the task of automatically assigning a proficiency score to a student’s essay written in response to a specific writing prompt, where the prompt defines the writing task and the topic the student is expected to address. Its importance lies in its ability to reduce the time, cost, and human effort required to grade essays, particularly in large-scale assessment settings. Beyond holistic scoring, which assigns a single overall score, trait scoring evaluates specific aspects of writing such as grammar, relevance, and vocabulary. Trait-level feedback is especially valuable in educational contexts, enabling students to identify strengths and weaknesses in their writing.

AES systems are typically developed under two paradigms. In prompt-specific AES, models are trained and evaluated on essays from the same prompt, enabling specialization in that prompt.

However, they often fail to generalize to essays written for unseen prompts, limiting practicality in real educational settings. In contrast, the cross-prompt setup trains models on essays from a set of source prompts and evaluates them on unseen target prompts, with the goal of improving generalization across prompts. A wide range of approaches has been proposed for cross-prompt AES, starting with feature-based methods (Li and Ng, 2024), followed by neural encoder-based models (Chen et al., 2025), and more recently, large language model (LLM)-based approaches (Yoshida, 2025). Although LLMs have demonstrated strong zero-shot capabilities across NLP tasks (Nourel-dien et al., 2025), existing AES studies suggest that their effectiveness is often limited when used in isolation (Mansour et al., 2024; Lee et al., 2024).

While most research has focused on English AES, Arabic AES is gaining momentum through new shared tasks and datasets (Bashendy et al., 2025, 2026). However, the progress has largely been limited to feature-based and neural approaches (Sayed et al., 2025; Albatarni et al., 2026), with little attention paid to LLMs. This gap is particularly notable given the recent emergence of Arabic-centric LLMs (Team et al., 2025; Bari et al., 2025) and the availability of large-scale Arabic AES benchmarks (Bashendy et al., 2026).

In this work, we investigate the potential of LLMs for Arabic AES. Specifically, we fine-tune three Arabic-centric LLMs (hereafter LLM^{ft}) for cross-prompt AES and evaluate them on the largest Arabic AES dataset, with the goal of providing a benchmark for Arabic-centric LLM performance. We aim to answer the following research questions:

- RQ1** Does LLM^{ft} improve performance compared to zero-shot and few-shot variants?
- RQ2** How does LLM^{ft} perform compared to state-of-the-art (SOTA) models?
- RQ3** How does LLM^{ft} perform in a cross-dataset setting compared to zero-shot and few-shot

prompting, and established AES baselines?

RQ4 What insights additional analyses provide into the reliability and scoring behavior of LLM^{ft}?

Our experiments show that LLM^{ft} significantly improves over its base variants and significantly outperforms SOTA baselines. Our contributions are three-fold:

- (1) We evaluate the performance of fine-tuning small Arabic-centric LLMs for Arabic cross-prompt AES, providing a systematic benchmark for their performance.
- (2) We provide a new SOTA for Arabic cross-prompt AES, offering a stronger point of comparison for future research.
- (3) We conduct extensive analyses, including cross-dataset evaluation and error analysis, providing insights into the behavior, generalization, and limitations of LLM^{ft}.

2 Related work

This section reviews prior work on AES, focusing on Arabic AES, followed by studies applying LLMs to AES through prompting and fine-tuning.

AES for Arabic Early Arabic AES research explored rule-based approaches and handcrafted linguistic features (Alqahtani and Alsaif, 2020; Alsanie et al., 2022), fine-tuning AraBERT (Ghazawi and Simpson, 2024), and applying parameter-efficient fine-tuning (Mahmoud et al., 2024). Sayed et al. (2025) pioneered cross-prompt AES for Arabic by introducing a large set of 816 handcrafted features and using them with various cross-prompt models. To advance research in Arabic AES, Bashendy et al. (2025) organized the TAQEEM shared task targeting cross-prompt scoring for Arabic AES, where participating teams primarily used AraBERT-based models and LLM prompting approaches. Most recently, Albatarni et al. (2026) proposed MAPLE, a cross-prompt AES model based on a meta-learning framework that leverages prototypical networks to learn transferable representations across different writing prompts, achieving SOTA performance on Arabic AES.

LLMs Prompting for AES When it comes to the utilization of LLMs for AES, most studies rely on prompting rather than fine-tuning. For example, Stahl et al. (2024) evaluated zero-shot, few-shot, and Chain-of-Thought prompting for essay scoring and feedback generation, while Mansour et al. (2024) explored prompt-engineering strate-

gies with ChatGPT and LLaMA. Similarly, Lee et al. (2024) proposed a zero-shot multi-trait specialization framework that decomposes the rubric into traits, for sequential evaluation. More advanced paradigms include reasoning-model comparisons (Yoshida, 2025) and a multi-agent evaluation framework (Jang et al., 2025). For Arabic AES, Mandhari et al. (2026) introduced a trait-specific framework that explored standard prompting, hybrid prompting with trait-specialist evaluators, and rubric-guided prompting under zero-shot and few-shot settings. Meanwhile, Qwaider et al. (2025) investigated prompt-essay relevance using conventional approaches such as threshold-based classifiers and SVMs, as well as prompting LLMs, including FalconH1, Qwen3, Gemma3, and Jais. Despite these innovations, prompting-based methods still fall short of SOTA AES systems in overall performance.

LLMs Finetuning for AES Few studies have explored fine-tuning LLMs for AES. Liu et al. (2025) fine-tuned GPT-3.5-turbo on TOEFL11 and observed an overrating tendency, though incorporating linguistic features improved the accuracy. Xiao et al. (2025) proposed a dual-process AES framework and showed that fine-tuning LLaMA 3 and GPT-3.5 can achieve performance comparable to conventional SOTA systems. In Arabic AES, Ghazawi and Simpson (2025) evaluated several fine-tuning strategies on ACEGPT and LLaMA, yet their models still underperformed compared to a smaller fine-tuned BERT-based model. Overall, while fine-tuning improves LLM performance, it often remains less competitive than smaller models. Our work addresses this gap by fine-tuning LLMs specifically for Arabic AES.

3 Study Design

In the cross-prompt AES setting, given a set of source prompts P_{src} , and a set of target prompts P_{trg} , the objective is to train a model on P_{src} such that it generalizes effectively to the *unseen* prompts P_{trg} . We aim to estimate the upper-bound performance of LLMs on cross-prompt AES by fine-tuning them for this task.

Multi-trait Scoring We fine-tune LLMs for trait-based essay scoring using a *Multi-Trait (MT)* setup, where a single model jointly predicts scores for multiple essay traits. This formulation allows the model to capture dependencies between traits while

enabling all traits to be predicted using a single model, improving both training and inference efficiency. Given a writing prompt and an essay, the model generates all trait scores in a structured JSON format. The fine-tuning prompt is shown in Figure 1.

سبتم إعطاءك مقال كتبه الطالب ردًا على الموضوع المعطى. مهمتك هي تقييم جميع السمات الكتابية ضمن النطاق المحدد. أخرج النتيجة بصيغة JSON فقط، دون أي شرح أو نص إضافي. يجب أن تكون درجة السمة رقمًا صحيحًا واحدًا داخل النطاق المحدد.

الموضوع: نص الموضوع الذي طلب من الطالب الكتابة عنه
المقال: نص المقال المكتوب من قبل الطالب ردًا على الموضوع
الدرجات: [" الصلة بالموضوع " : 0-2 ، " الهيكل العام " : 0-5 ، " المفردات " : 0-5 ، " الأسلوب والتناسك البنائي " : 0-5 ، " الأفكار والمضمون " : 0-5 ، " الإملاء والترقيم والشكل " : 0-5 ، " البناء والتراكيب " : 0-5 ، " التقييم الكلي " : 0-32]

{prompt} : الموضوع
{essay} : المقال
الدرجات:

Figure 1: The Arabic prompt used for multi-trait prediction. English translation is provided in Figure 2.

You will be given an essay written by a student in response to a prompt. Your task is to evaluate all the writing traits within the specified range.
Output the result in JSON format only, without any additional explanation or text. Each trait’s score should be an integer within the specified range.

—

Prompt: The prompt that the student was asked to write about
Essay: The essay written by the student in response to the prompt
Scores: { ‘Relevance’: (0, 2), ‘Organization’: (0, 5), ‘Style’: (0, 5), ‘Development’: (0, 5), ‘Mechanics’: (0, 5), ‘Grammar’: (0, 5), ‘holistic’: (0, 32) }

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Prompt: {prompt}
Essay: {essay}
Scores:

Figure 2: English translation of the Multi-trait prediction prompt. The Arabic version is shown in Figure 1.

Fine-Tuning Setting We fine-tune the LLMs using the Hugging Face Transformers library with the PEFT¹ framework for parameter-efficient adaptation. Specifically, we apply LoRA to the causal language modeling objective, keeping the base model frozen while training low-rank adapters (rank $r = 64$, $\alpha = 128$, dropout = 0.1). The adapters are applied to the attention projection matrices (Key, Query, Value, and Output) and MLP

layers (Gate, Up, and Down projections). Supervised fine-tuning is conducted using the TRL library’s SFTTrainer,² with AdamW optimizer, a learning rate of 2×10^{-4} , and a cosine scheduler. These hyperparameter values were determined based on preliminary experiments. Moreover, training uses a micro-batch size of 4 with gradient accumulation steps of 16 (global batch size 64), for 10 epochs with a maximum sequence length of 2048. Mixed-precision training with bfloat16 is enabled for computational efficiency. For inference, we use greedy decoding for all experiments. The best checkpoint is selected based on performance on the development set. All experiments were run using one H200 GPU.

Dataset We used LAILA (Bashendy et al., 2026), a large-scale Arabic AES dataset with 7,859 essays written in response to eight prompts (P1-P8). Essays were annotated for seven traits: relevance (REL) on a scale of 0–2, and organization (ORG), vocabulary (VOC), style (STY), development (DEV), mechanics (MEC), and grammar (GRA) on a scale of 0–5, and a holistic (HOL) score as the sum of all trait scores. We adopt the cross-prompt splits provided by LAILA authors.³

Fine-tuned LLMs For fine-tuning, we select small open-source Arabic-centric LLMs. We focus on models in the 7B–9B parameter range, as they are practical to fine-tune while remaining computationally efficient. Moreover, to ensure architectural diversity, we choose models built on different base architectures. Specifically, we finetune FANAR 9B (Team et al., 2025),⁴ ALLAM 7B (Bari et al., 2025),⁵ and JAIS 8B (Anwar et al., 2025).⁶

Other Scoring Models We compare the fine-tuned LLMs against a range of prompting-based LLM baselines and SOTA engineered models:

1. **Zero-shot (LLM⁰):** We evaluate the 3 fine-tuned LLMs (FANAR 9B, ALLAM 7B, and JAIS 8B) in a zero-shot setting to assess their inherent scoring capabilities without task-specific adaptation. We additionally evaluate larger variants, namely FANAR 27B and JAIS

²https://huggingface.co/docs/trl/en/sft_trainer

³<https://gitlab.com/bigirqu/laila>

⁴<https://huggingface.co/QCRI/Fanar-1-9B>

⁵<https://huggingface.co/humain-ai/ALLaM-7B-Instruct-preview>

⁶<https://huggingface.co/inceptionai/Jais-2-8B-Chat>

¹<https://huggingface.co/docs/peft/en/index>

Model	Size	REL	ORG	VOC	STY	DEV	MEC	GRA	HOL	AVE
ALLAM ⁰	7B	0.230	0.388	0.416	0.463	0.393	0.415	0.434	0.377	0.390
JAIS ⁰	8B	0.112	0.126	0.169	0.176	0.153	0.215	0.211	0.234	0.175
JAIS ⁰	70B	0.112	0.334	0.431	0.367	0.413	0.433	0.443	0.461	0.374
FANAR ⁰	9B	0.457	0.536	0.421	0.550	0.487	0.371	0.448	0.521	0.474
FANAR ⁰	27B	0.040	0.459	0.262	0.084	0.261	0.175	0.095	0.427	0.225
GPT-5 ⁰	-	0.620	0.709	0.540	0.586	0.652	0.410	0.547	0.716	0.598
ALLAM ^{fs}	7B	0.096	0.253	0.313	0.316	0.273	0.350	0.339	0.347	0.286
JAIS ^{fs}	8B	0.093	0.226	0.277	0.278	0.255	0.260	0.261	0.304	0.244
FANAR ^{fs}	9B	0.425	0.591	0.552	0.607	0.563	0.582	0.594	0.651	0.570
MOOSE	0.6B	0.411	0.627	0.642	0.649	0.585	0.586	0.623	0.649	0.597
XGB	-	0.360	0.645	0.641	0.641	0.583	0.577	0.619	0.679	0.593
MAPLE	0.16B	0.290	0.647	0.686	0.702	0.607	0.634	0.664	0.723	0.619
TRATES	-	0.557	0.696	0.657	0.664	0.652	0.608	0.643	0.744	0.653
ALLAM ^{ft}	7B	0.654*•	0.771*•	0.731*	0.741*	0.716	0.696*•	0.708*•	0.781*•	0.725*•
JAIS ^{ft}	8B	0.552	0.743*	0.708*	0.714	0.681	0.659	0.678	0.746	0.685
FANAR ^{ft}	9B	0.656*•	0.789*•	0.743*	0.752*•	0.718*•	0.714*•	0.726*•	0.796*•	0.737*•

Table 1: Performance of LLM^{ft} against the other scoring models on LAILA test split. Statistical significance is done using a t-test ($p < 0.05$) over results across individual prompts. Improvements over TRATES are denoted by *. Improvements over JAIS^{ft} are denoted by •. FANAR^{ft} and ALLAM^{ft} are not statistically significant.

70B. We also include GPT-5 as a reference for large frontier models.

2. **Few-shot (LLM^{fs}):** Similarly, we evaluate the 3 LLMs under few-shot scoring using 5 in-context examples randomly sampled from different prompts in P_{src} , to assess the extent to which task demonstrations improve scoring performance. Since few-shot performance can be sensitive to the selected examples, we repeat each experiment three times with different random seeds and report the average performance.
3. **SOTA:** We select 4 engineered models that achieve the best performance on LAILA. MOOSE is an adaptation of an English cross-prompt AES model based on a mixture-of-experts architecture (Chen et al., 2025; Bashendy et al., 2026). XGB is a strong feature-based model reported in (Bashendy et al., 2026). MAPLE is a meta-learning framework for cross-prompt AES that incorporates prompt, rubric, and generic features into the essay representation (Albatarni et al., 2026). TRATES is a system that generates trait-specific features and combines them with linguistic features for scoring using a simple NN (Eltanbouly et al., 2025). We use its Arabic adaptation by Elbahnasawi et al. (2026), which is the current SOTA on LAILA.

The LLM experiments were run using vLLM (Kwon et al., 2023) and relevant API’s for closed models. For smaller LLMs, we use *structured outputs*, since the smaller models are not able to adhere to the output format consistently.

Evaluation Metric We evaluate our models using Quadratic Weighted Kappa (QWK) (Cohen, 1968), a widely used metric in AES. QWK quantifies agreement between human and system scores while accounting for the severity of disagreements, making it suitable for ordinal score scales.

4 Experimental Evaluation

This section addresses the first three research questions: LLM^{ft} vs. Non-finetuned LLMs, LLM^{ft} vs. SOTA, and cross-dataset performance of LLM^{ft}.

4.1 LLM^{ft} vs. Non-tuned LLMs (RQ1)

Table 1 compares LLM^{ft} performance with the other scoring models on LAILA test set.⁷

Non-tuned LLMs For the prompting-based LLMs, FANAR^{fs} and JAIS^{fs} perform better than their LLM⁰ variants, with improvements of 10 and 7 points, respectively. Notably, the larger model, JAIS⁰ 70B, outperforms JAIS^{fs} 8B by 13 points. In contrast, ALLAM shows the opposite trend, where few-shot prompting leads to a performance drop of

⁷Per-prompt results are in Table 4.

almost 10 points. Overall, the results of non-tuned LLMs are underwhelming, not surpassing existing SOTA models or the frontier model, GPT-5⁰.

Fine-tuning Fine-tuning leads to substantial performance improvements, with gains of 17, 34, and 44 points for FANAR^{ft} , ALLAM^{ft} , and JAIS^{ft} , respectively, over their best non-tuned configurations. Interestingly, although the differences in performance between LLM⁰ models and also between the LLM^{fs} models were large (e.g., FANAR^0 outperforming ALLAM^0 and JAIS^0 by 8 and 30 points, respectively), those gaps narrow significantly after fine-tuning, with FANAR^{ft} exceeding ALLAM^{ft} and JAIS^{ft} by only 1 and 5 points, respectively.

4.2 LLM^{ft} vs SOTA Models (RQ2)

Table 1 shows that LLM^{ft} outperforms all SOTA models by a substantial margin. In fact, the best-performing LLM, FANAR^{ft} , outperforms the best baseline model, TRATES, by 8 points, establishing a new SOTA performance in Arabic AES on LAILA dataset. Moreover, improvements of FANAR^{ft} , ALLAM^{ft} , and JAIS^{ft} over TRATES are statistically significant in 8, 7, and 2 traits, respectively, with both FANAR^{ft} and ALLAM^{ft} also showing statistically significant improvements on average. Notably, differences in performance between FANAR^{ft} and ALLAM^{ft} are not statistically significant across any individual trait, indicating comparable performance between the two models. In contrast, improvements of FANAR^{ft} over JAIS^{ft} are statistically significant in all traits except VOC.

4.3 LLM^{ft} in Cross-dataset Setting (RQ3)

We now turn to evaluating LLM^{ft} in a cross-dataset setting. This setting assesses the generalizability of AES models not only across unseen prompts but also across different essay types and demographics (Eltanbouly et al., 2026). It is particularly relevant in low-resource scenarios, where small datasets are insufficient for training billion-parameter LLMs.

Dataset For this experiment, we use the ZAE-BUC dataset (Habash and Palfreyman, 2022). It contains 214 Arabic essays written by first-year university students across three topics. While ZAE-BUC differs from LAILA in prompts, essay characteristics, and annotation scheme, the two populations are expected to have broadly comparable language proficiency, as LAILA targets high school students, including those in their final year, who are

Model	Size	QWK	RMSE
ALLAM^0	7B	0.267	1.189
JAIS^0	8B	0.160	1.187
FANAR^0	9B	0.451	0.877
ALLAM^{fs}	7B	0.470	0.772
JAIS^{fs}	8B	0.333	1.363
FANAR^{fs}	9B	0.306	0.972
RF	-	0.477	0.639
XGB	-	0.427	0.838
MOOSE	0.6B	0.329	1.083
ALLAM^{ft}	7B	0.481	0.697
JAIS^{ft}	8B	0.444	0.710
FANAR^{ft}	9B	0.533	0.628

Table 2: Cross-dataset performance of LLM^{ft} and baseline systems (LAILA → ZAEBUC)

at a similar educational stage to first-year university students. The dataset provides CEFR language proficiency annotations, which we map to scores in the range [1–6], following Habash and Palfreyman (2022). For predictions, we map LAILA’s holistic scores from the original [0–32] range to the [1–6] range used by ZAEBUC. Given the small size of ZAEBUC, we report QWK over the entire dataset rather than computing prompt-level scores and averaging them. Additionally, we include RMSE as a complementary metric, as QWK can exhibit instability on small datasets (Albatarni and Elsayed, 2026).

Scoring Models We evaluate the following models on ZAEBUC:

1. **LLM⁰**: Since ZAEBUC contains only the overall score of the essays, we prompt the same 3 LLMs (FANAR, ALLAM, and JAIS) to predict the holistic score (prompt depicted in Figure 6), with a score range of 1–6.
2. **LLM^{fs}**: We use the same prompt as LLM⁰, augmented with 5 random examples from LAILA. The examples include holistic scores only. Since the holistic score range in LAILA is 0–32, we prompt the LLM to generate scores in this range, then map the predictions to ZAEBUC’s score range. We repeat the experiment three times with different random seeds and report the average performance.
3. **Qayyem’s Models**: We consider the models available in Qayyem,⁸ an online Arabic AES platform (Elbahnasawi et al., 2026) namely RF, XGB, and MOOSE.

⁸At the time of the experiments

نص الموضوع: بات إهتمام وحماس المراهقين لتعلّم رياضة جديدة أو الانتظام في حضور التّدريبات الرياضية الخاصة بها يتّضاءل يوماً بعد يوم؛ حتّى صار يُدق ناقوس خطرٍ مقلق يُنذر بوجود جيل ضَعيف البنية (الجسم). اكتب مقالاً مكوناً من ثلاثمائة (٣٠٠) كلمة توضّح فيه أسباب انتشار هذه الظاهرة، مُراعياً سمات المقال التفسيري، وسلامة اللغة، وموطئاً علامات الترقيم وأدوات الربط بشكلٍ صحيح.

Prompt (English translation): The interest and enthusiasm of teenagers to learn a new sport or regularly attend its training sessions is dwindling day by day, to the point that it is sounding an alarming warning of a generation with weak physiques. Write an essay of 300 words explaining the reasons for the spread of this phenomenon, observing the characteristics of an expository essay, maintaining correct grammar, and employing punctuation and conjunctions correctly.

الموضوع: تزداد السمنة يوم بعد يوم في قطر وذلك لاهمال الافراد لصحتهم وعدم ممارستهم للرياضة وقضاء اغلب اوقاتهم في المنزل وعدم استغلالهم للوقت وعمل أنشطة رياضية. ارى ان هناك عدة اسباب لهذه المشكلة ومنها ان العائلة قد تهمل احيانا الرياضة ولا تمانعون في جلوس ابنائهم في البيت طوال الوقت وعدم الحركة ولكن يجب على الاهالي تخصيص وقت للرياضة او تسجيل ابنهم في اندية رياضية فنحن في قطر تتوفر لدينا العديد من الاندية للانشطة الرياضية وهي نوادي الرياضة او الاندية الكروية او السباحة او حتى تعلم فنون القتال، كل هذه الاندية يمكن للاطفال الانضمام لها وتخصيص وقت يوميا او حتى تخصيص اربع ايام في الاسبوع لهذه الانشطة وتسجيل كل ابن في ما يحب للاستمتاع والفائدة من حيث عمله لحركات رياضية. ايضا اصبحت بعض المدارس تهمل الحصص الرياضية لانهم يعتقدون بأن المواد الاخرى اكثر اهمية ولكن هذا خطأ ويجب على المدارس ايضا تخصيص أنشطة رياضية للطلاب ويمكن ذلك عن طريق عمل عدة أنشطة في وقت الفسحة وجعل الطلاب يشاركون.

الموضوع: أن المراهقين اليوم أصبحوا يميلون إلى الأشياء الخطرة عليهم مما أدى إلى ترك الرياضة والاكتفاء بالراحة لم يدركوا بأن الراحة لا تكمل إلى بالرياضة فراحة الجسم والبدن تكمل بتنوع الرياضة مثل التنس والجري وبكرة القدم وغيرها من ذلك فأصبحوا يروون بعدم وجود أهمية الرياضة كونها ليست أساسية في الحياة فيجب على كل مراهق بأن يدرك خطورة التهاون في الجسم وسلامة فسلامة الجسم تكمل باستخدام الرياضة كافة بأنواعها والالتزام به يجب عليهم الذهاب البطولات حتى يتشجع من خلاله أو حضور مباراة أو دوريات عالية حتى يتشجع ويبادر من نفسه الصحة جسمه من الأمراض مستقبلاً ويجب الالتزام بالتدريبات حتى يتمكن من الاستفادة من ذلك فلا يركز الفرد على شيء معين في الرياضة حتى لا يتملل من ذلك وترك الرغبة بالالتزام فيه يختص يومين فالاسبوع على الأقل الذهاب حتى لو يوم فالرياضة في يومنا هذا مهم ومهم جداً يشارك في البطولات والاندية المدرسية حتى يتمكن من الاستفادة بها وفيها أيضاً فترك الشيء يعني أنعدامه في حياة كل مراهق ومراهقة اليوم وغداً

Figure 3: Two essays from Prompt 1 with a true relevance score of 1. The left essay (011026) is **correctly** scored as 1, while the right essay (010926) is **incorrectly** scored as 2.

4. LLM^{ft} : We train LLM^{ft} on all prompts in LAILA, using the most frequent number of steps across the individual folds for each LLM, and then evaluate the resulting model on ZAEBUC. LLM^{ft} predicts all LAILA traits; however, we consider only HOL prediction and map the scores to ZAEBUC's scale.

Evaluation Results The results are presented in Table 2. Overall, the performance of LLM^0 on ZAEBUC closely mirrors its behavior on LAILA dataset, where ALLAM and FANAR consistently outperform JAIS. Notably, although JAIS⁰ shows lower performance in terms of QWK, RMSE indicates that its errors are comparable to ALLAM⁰, with only a small difference. LLM^{fs} prompting improves the performance of ALLAM and JAIS by 20 and 17 points, respectively, whereas FANAR experiences a 15-point decline. This suggests that few-shot examples from another dataset can benefit some LLMs but may also introduce instability for others.

In contrast, fine-tuning on LAILA leads to substantial performance gains across all models. Specifically, JAIS^{ft} and FANAR^{ft} improve by 11 and 8 points, respectively, over their best non-tuned configurations, with consistent improvements also observed in RMSE. ALLAM^{ft}, however, improves by only 1 QWK point. This demonstrates that supervised fine-tuning on essay data enhances cross-dataset generalization, even when the training and test sets differ in demographics and topics. Importantly,

direct fine-tuning on ZAEBUC would not be practical due to its extremely small size.

Finally, when compared against the engineered AES systems, FANAR^{ft} outperforms the strongest model (RF) by 6 QWK points while achieving comparable RMSE. Additionally, ALLAM^{ft} achieves performance comparable to RF on both QWK and RMSE. Unlike the substantial gains observed on LAILA, improvements in the cross-dataset setting are more modest, underscoring the continued challenge of generalizing across datasets.

5 Performance Analysis

In this section, we address the final research question: what do additional analyses reveal about the reliability and scoring behavior of LLM^{ft} ? We conduct a detailed analysis on the best-performing model, FANAR^{ft}, and investigate the following sub-research questions:

1. How do errors in the REL trait affect the prediction of other traits? (§5.1)
2. What are the underlying causes of errors in the REL trait, as observed through qualitative case study analysis? (§5.2)
3. Does single-trait fine-tuning yield better performance than multi-trait fine-tuning? (§5.3)
4. To what extent is the multi-trait model influenced by autoregressive bias during generation? (§5.4)

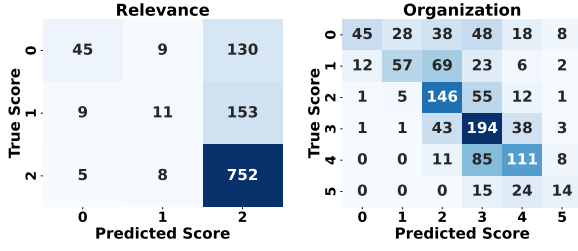


Figure 4: REL and ORG traits confusion matrix for P1.

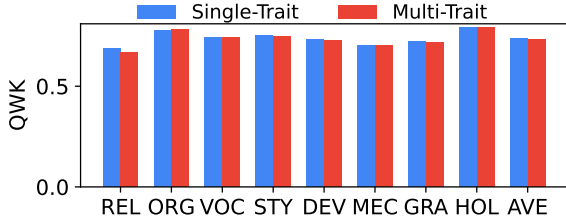


Figure 5: LLM^{ft} results with ST and MT models.

5.1 Error Analysis

We begin by examining writing prompt P1, which is the most challenging prompt, with an average QWK of 0.488, compared to the other prompts, with QWK ranging from 0.733 to 0.805 (see Appendix A). To understand the source of this performance gap, we examine the REL trait, which obtains the lowest QWK for P1 (0.303). We then analyze the ORG trait to investigate how errors in REL prediction propagate to the scoring of other traits. Figure 4 shows the confusion matrix for REL and ORG traits, explaining this weaker performance. In LAILA, essays with a REL score of zero automatically receive zeros for all other traits, and P1 contains a relatively high proportion of such essays (16.4%). Our analysis shows that although the model captures this dependency (assigning zero to other traits when REL is predicted as zero), it struggles to predict REL accurately. For example, 130 essays with a true REL of 0 were predicted as 2, causing excessive over-predictions in the other traits. Specifically, 112 essays of them received a score above 1 for ORG, contributing to the lower overall performance.

Another interesting observation highlights that across all prompts, only 8.1% of essays received the same scores as the gold labels across all traits. However, 41.6% of essays are predicted within a one-point difference for all traits, suggesting that while exact matches are relatively limited, the model’s predictions are pretty close to the true scores, with an average RMSE of 1.123 across all traits.

5.2 Case Study

To further investigate the REL issue, we examine two essays: one correctly predicted and one misclassified. Figure 3 presents two essays from P1 with a true REL score of 1. The model correctly predicts the first essay (left), which exhibits a clear topical mismatch with the prompt, focusing on obesity and general health-related issues rather than teenagers’ declining interest in sports participation. However, the second essay (right) is overpredicted as 2. While it contains several prompt-related keywords (e.g., teenagers, sports, training, and competitions), it provides limited discussion of the prompt’s main question and lacks sufficient explanatory depth. This example suggests that the model relies heavily on lexical overlap when assessing REL. As a result, essays containing prompt-related keywords may receive inflated scores despite only partially addressing the prompt.

5.3 ST vs. MT

سيتم إعطاؤك مقال كتبه الطالب ردًا على الموضوع المعطى. مهمتك هي تقييم سمة التقييم الكلي ضمن النطاق (٠-٣٢). أخرج النتيجة بصيغة جصو فقط، دون أي شرح أو نص إضافي. يجب أن تكون درجة السمة رقمًا صحيحًا واحدًا داخل النطاق المحدد. -- الموضوع: نص الموضوع الذي طلب من الطالب الكتابة عنه المقال: نص المقال المكتوب من قبل الطالب ردًا على الموضوع الدرجات: (أنتقيم الكلي: 0-32) -- {prompt}: الموضوع {essay}: المقال الدرجات:
--

Figure 6: The Arabic prompt used for the ST prediction. English translation is provided in Figure 7.

We next investigate whether fine-tuning a separate model for each trait, referred to as **Single-Trait (ST)**, yields better performance than MT. For this experiment, we use FANAR. Under the ST setting, a dedicated model is fine-tuned to predict a single trait score for a given essay, resulting in one model per trait (the prompt is depicted in Figure 6).

The results on the development set are shown in Figure 5. Interestingly, the average performance of ST and MT models is very close. REL is the only trait with a noticeable difference, where ST outperforms MT by about 2 points. This may be because REL is scored on a smaller range (0-2) than the other traits (0-5), so MT training may dilute its gradient signal. Despite this small difference, the MT

You will be given an essay written by a student in response to a prompt. Your task is to evaluate the holistic trait within the range (0, 32). Output the result in JSON format only, without any additional explanation or text. The score should be an integer within the specified range. Prompt: The prompt that the student was asked to write about Essay: The essay written by the student in response to the prompt Score: { 'Holistic': (0, 32) } Prompt: {prompt} Essay: {essay} Score:
--

Figure 7: English translation of the ST prediction prompt.

model is more efficient, predicting all eight scores with one model, while the ST approach requires eight separate models. Specifically, the inference time of the ST approach is 0.35 seconds per essay for all traits, compared to 0.07 seconds for the MT model. Training the ST models for all traits over 10 epochs requires approximately 678 hours, compared to about 102 hours for the MT model.

5.4 Autoregressive Copying

We also observed a tendency of the MT model to repeat scores across traits. On the test set, 45.7% of essays have identical scores for all traits (excluding REL and HOL), compared to 15.2% in the gold labels. This raises a concern: “is the MT model biased by its autoregressive generation process?” To investigate this concern, we compare MT and ST predictions on the development set. For each consecutive trait pair $(i, i + 1)$, we focus on cases where the *gold scores differ* ($y_i \neq y_{i+1}$) and define two probabilities:

$$P_{\text{SEQ}} = p(\hat{y}_{i+1}^M = \hat{y}_i^M \text{ and } \hat{y}_{i+1}^S \neq \hat{y}_i^S \mid y_{i+1} \neq y_i)$$

$$P_{\text{IND}} = p(\hat{y}_{i+1}^M = \hat{y}_i^M \text{ and } \hat{y}_{i+1}^S = \hat{y}_i^S \mid y_{i+1} \neq y_i)$$

where $\hat{y}^S, \hat{y}^M$ are the predicted scores by ST and MT models, respectively. P_{SEQ} measures the probability that MT introduces copying between trait scores that ST distinguishes, suggesting *sequential* influence, while P_{IND} captures cases where the MT model’s repetition aligns with the ST approach’s *independent* assessment.

As shown in Table 3, P_{IND} is significantly higher than P_{SEQ} across all pairs (averaging 0.526 vs. 0.215). This indicates that repeated scores in MT

trait_i	trait_{i+1}	P_{SEQ}	P_{IND}
ORG	VOC	0.152	0.377
VOC	STY	0.217	0.476
STY	DEV	0.243	0.598
DEV	MEC	0.249	0.432
MEC	GRA	0.214	0.750
AVE		0.215	0.526

Table 3: Consecutive trait copying.

are not primarily due to autoregressive propagation; rather, the MT model assigns these scores deliberately, reflecting a consistent evaluation process similar to the independent ST predictions.

6 Conclusion and Future Work

This study evaluated the effectiveness of fine-tuning relatively small Arabic-centric LLMs, namely ALLAM 7B, JAIS 8B, and FANAR 9B, for cross-prompt Arabic automated essay scoring. The results show that LLM^{ft} consistently outperforms strong baselines, including much larger proprietary models such as GPT-5. Specifically, the best-performing model, FANAR^{ft} , surpasses the strongest baseline on LAILA dataset by 8 points. Moreover, we evaluate LLM^{ft} in a cross-dataset setting (training on LAILA and testing on ZAE-BUQ), where FANAR^{ft} achieves the strongest performance among all models. Further analysis of FANAR^{ft} reveals several performance patterns, highlighting both its strengths and weaknesses. In future work, we plan to explore alternative prompting strategies, incorporate rubric-based information, and investigate model distillation to improve efficiency while maintaining strong performance.

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7 Limitations

While our results are substantial, this study has several limitations. First, we evaluated only two prompt formulations and did not explore a broader

range of prompting strategies that may further improve performance. Second, our benchmark focuses on fine-tuning relatively small Arabic-centric LLMs and does not investigate larger or multilingual LLMs. Future work could examine whether the observed gains transfer to these settings and scale with model size and language diversity. Finally, fine-tuning LLMs requires non-trivial computational resources, which may limit accessibility in some settings.

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A Per-Prompt Results

We present the per-prompt results on LAILA dataset in Table 4. Prompt P1 shows the lowest performance, with an average QWK of 0.488, compared to the other prompts, whose performance ranges from 0.733 to 0.805.

LLM	Prompt	REL	ORG	VOC	STY	DEV	MEC	GRA	HOL	AVE
ALLAM ^{ft}	P1	0.373	0.568	0.547	0.553	0.531	0.504	0.514	0.553	0.518
	P2	0.573	0.757	0.730	0.733	0.699	0.658	0.683	0.756	0.699
	P3	0.710	0.876	0.782	0.798	0.755	0.786	0.812	0.868	0.798
	P4	0.690	0.815	0.759	0.769	0.731	0.764	0.755	0.813	0.762
	P5	0.625	0.774	0.691	0.729	0.728	0.679	0.713	0.775	0.714
	P6	0.683	0.782	0.762	0.758	0.752	0.732	0.757	0.814	0.755
	P7	0.699	0.811	0.792	0.755	0.712	0.742	0.756	0.838	0.763
	P8	0.757	0.817	0.835	0.812	0.740	0.772	0.764	0.854	0.794
JAIS ^{ft}	P1	0.235	0.524	0.502	0.525	0.458	0.471	0.473	0.500	0.461
	P2	0.334	0.731	0.709	0.745	0.670	0.657	0.675	0.741	0.658
	P3	0.654	0.830	0.791	0.751	0.709	0.748	0.787	0.826	0.762
	P4	0.599	0.814	0.739	0.771	0.722	0.751	0.747	0.800	0.743
	P5	0.614	0.761	0.683	0.729	0.732	0.677	0.714	0.771	0.710
	P6	0.625	0.761	0.756	0.742	0.722	0.706	0.741	0.799	0.732
	P7	0.685	0.802	0.808	0.739	0.742	0.711	0.731	0.829	0.756
	P8	0.521	0.740	0.737	0.716	0.648	0.647	0.662	0.738	0.676
FANAR ^{ft}	P1	0.303	0.548	0.505	0.545	0.502	0.482	0.494	0.522	0.488
	P2	0.731	0.815	0.793	0.786	0.772	0.740	0.756	0.835	0.779
	P3	0.697	0.885	0.772	0.823	0.762	0.796	0.813	0.874	0.803
	P4	0.619	0.837	0.742	0.788	0.736	0.757	0.750	0.811	0.755
	P5	0.647	0.784	0.724	0.734	0.749	0.697	0.734	0.793	0.733
	P6	0.701	0.783	0.756	0.771	0.733	0.742	0.761	0.820	0.759
	P7	0.751	0.829	0.808	0.752	0.718	0.737	0.746	0.847	0.774
	P8	0.799	0.831	0.842	0.820	0.772	0.759	0.753	0.866	0.805

Table 4: Per prompt and per trait QWK performance on LAILA test set